



COMMENTARY

Driving Best Practices Throughout the Treatment Journey for Patients with NSCLC with Actionable Alterations: A Podcast Discussion

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ABSTRACT

Non-small cell lung cancer (NSCLC) treatment has been revolutionized by the advent of targeted therapies for tumors harboring specific actionable alterations. Targeted agents are now approved for use in patients with advanced NSCLC with various drivers including *ALK* rearrangements, *BRAF* V600E mutations, *EGFR* mutations, *ERBB2* mutations, *KRAS* G12C mutations, *MET* exon 14 skipping alterations, *NTRK* fusions, *RET* rearrangements, and *ROS1* rearrangements. Importantly, the availability of these agents has raised the clinical question of how to optimally sequence their use alongside chemotherapy and/

or immunotherapy strategies, which are indicated for broader populations. Key considerations include (i) evidence for better outcomes when first-line treatment is initiated following availability of molecular profiling data; (ii) the decreasing proportion of patients able to receive therapy in each successive treatment line; (iii) the efficacy of targeted agents demonstrated in either single-arm trials or head-to-head comparisons with chemotherapy and/or immunotherapy, as compared with evidence for poor or modest efficacy of immunotherapy in patients with tumors with actionable drivers; (iv) real-world data showing better outcomes of patients with tumors with actionable alterations who received targeted therapies compared with those who did not; (v) the generally favorable safety profile of targeted therapies, as well as the potential for increased toxicity when immunotherapy precedes certain targeted agents; and (vi) patient-centric factors including the greater ease of administration of oral targeted therapies over intravenous chemotherapy or immunotherapy strategies. In line with these considerations, guidelines typically recommend most targeted agents approved for first-line use as initial therapy over chemotherapy and/or immunotherapy. In this podcast, the authors discuss the current therapeutic landscape of NSCLC with actionable alterations and provide their perspectives on treatment algorithms, and how to optimally sequence therapies for patients with tumors

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harboring actionable alterations, using patient cases to illustrate key principles.

PLAIN LANGUAGE SUMMARY

Several targeted therapies are now available for the treatment of patients with advanced non-small cell lung cancer that has certain genetic alterations. These medicines specifically target the genetic alterations in cancer cells that increase their ability to grow and spread. Targeted therapies are approved for patients whose tumors have changes in genes such as *ALK*, *BRAF*, *EGFR*, and others. In this podcast, three oncologists engage in a discussion on the optimal use of targeted therapies alongside other treatments like chemotherapy and immunotherapy. Their discussion highlights the importance of analyzing the tumor's genetic makeup before treatment to determine if targeted therapy is an option. Cancer guidelines typically recommend using targeted therapies as the first treatment option in patients with non-small cell lung cancer with specific genetic alterations, if approved for this use. Since many patients will

receive only one therapy, informed selection of the initial treatment is especially important. The authors also consider the data from clinical trials showing the effectiveness and manageable side-effect profile of targeted therapies. In contrast, there is evidence that immunotherapies may be less effective in patients whose tumors have certain genetic alterations. Real-world data support the importance of using targeted therapies in patients who are eligible. In addition, targeted therapies are often taken as oral tablets, which may be preferred by patients to intravenous treatment with chemotherapy or immunotherapy. Finally, the authors use example patient cases to show important factors to consider when choosing between targeted therapies and other types of treatment.

Keywords: Non-small cell lung cancer; Treatment sequencing; Actionable biomarkers; *MET* exon 14 skipping; Biomarker testing; Targeted therapy

Infographic

Driving best practices throughout the treatment journey for patients with NSCLC with actionable alterations: A podcast discussion

Christine M. Bestvina, Chul Kim, Nathalie Daaboul

Current landscape of actionable alterations in patients with advanced NSCLC

Actionable alterations with approved targeted therapies^{1,2}

ALK rearrangements	BRAF V600E mutations	EGFR mutations
ERBB2 mutations	KRAS G12C mutations	MET exon 14 skipping alterations
NRG1 gene fusion	NTRK fusions	RET rearrangements
		ROS1 rearrangements

Importance of biomarker testing²

Broad biomarker testing is essential irrespective of patient characteristics such as age, sex, or smoking history to identify candidates for targeted therapies

Treatment options for patients with NSCLC with actionable alterations²



Targeted therapies
(mostly oral)



Platinum-based
chemotherapy



Immunotherapy
(with or without chemotherapy)



How to optimally sequence targeted therapies alongside chemotherapy and/or immunotherapy

Clinical considerations for treatment sequencing



Potential delays in biomarker testing

First-line treatment should be initiated following receipt of genomic data whenever feasible^{1,4}



Limited number of treatment lines

Since many patients receive only a single line of therapy, they may not receive effective therapies if deferred for later use^{1,4}



Clinical trials

Approved targeted therapies generally show better outcomes than chemotherapy^{1,5}; some evidence suggests poor/modest efficacy of immunotherapy in patients with tumors with actionable drivers⁶



Real-world evidence

Comparisons based on real-world data showed better outcomes with targeted therapies compared with other treatments^{6–11}

Safety



Adverse-event profiles differ between classes of targeted agents and can include peripheral edema, diarrhea, and nausea^{12–14}

Common adverse events associated with targeted therapies are generally manageable^{15–17}

Toxicity of some targeted therapies can be exacerbated by prior immunotherapy^{16–18}



Patient-centric considerations

Chemotherapy and immunotherapies require regular clinic visits for intravenous treatment administration⁷

Orally administered targeted therapies may be more convenient options for patients

Some targeted therapies have shown improvements in patients' symptoms and quality of life compared with chemotherapy^{19–22}



Guidelines recommend most targeted agents approved for first-line use as initial therapy over chemotherapy and/or immunotherapy for patients whose tumors have corresponding alterations^{2,23}

¹Based on data from single-arm trials.

References: 1. Anaghi M, et al. *Cancer Cell Int*. 2023;23:162. 2. Referenced with permission from the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) for Non-Small Cell Lung Cancer V.3.2025. © National Comprehensive Cancer Network, Inc. 2025. All rights reserved. Accessed [January 17, 2025]. To view the most recent and complete version of the guideline, go online to NCCN.org. NCCN makes no warranties of any kind whatsoever regarding their content, use or application and disclaims any responsibility for their application or use in any way. 3. Aggarwal C, et al. *JCO Precis Oncol*. 2023;7:e23010191. 4. Scott JA, et al. *JCO Oncol Pract*. 2004;20:145–53. 5. Maini-Akvedo JA, et al. *Cancers (Basel)*. 2021;15:620. 6. Blasi M, et al. *Eur J Cancer*. 2024;199:113556. 7. Phillips WJ, et al. *CMAJ*. 2024;196:E558–61. 8. Mazieres J, et al. *Ann Oncol*. 2019;30:1321–8. 9. Singal G, et al. *JAMA*. 2019;321:1391–9. 10. Yang M, et al. *Cancer Med*. 2023;12:21605–14. 11. Zhao S, et al. *BMC Med*. 2021;19:223. 12. Hsu WH, et al. *Ann Oncol*. 2018;29:13–9. 13. Liguori V, et al. *Biomedicines*. 2023;11:2538. 14. Cortot A, et al. *Clin Lung Cancer*. 2022;23:195–207. 15. Mitchell CL, et al. *Diagnostics (Basel)*. 2023;13:1117. 16. Schoenfeld AJ, et al. *Ann Oncol*. 2019;30:839–44. 17. Desai A, et al. *Cancer Treat Res Commun*. 2023;36:100743. 18. Chour A, et al. *J Thorac Oncol*. 2023;18:1408–15. 19. Chan BA, Hughes BGH. *Transl Lung Cancer Res*. 2015;4:36–54. 20. Cameron LB, et al. *Cochrane Database Syst Rev*. 2022;1:CD013453. 21. Zhou C, et al. *N Engl J Med*. 2023;389:1839–50. 22. Billings NE, et al. *Cancers (Basel)*. 2021;13:4282. 23. Jayesim JA, et al. *J Clin Oncol*. 2024;42:E1–22.

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PEER-REVIEWED
FEATURE



Key Summary Points

The approval of numerous targeted therapies for patients with advanced non-small cell lung cancer with corresponding driver alterations has raised the clinical question of how to best sequence these therapies alongside chemotherapy and/or immunotherapy, which are used in broader populations.

This podcast explores the key considerations for treatment sequencing for these patients, including potential delays in biomarker testing, the limited ability to administer multiple treatment lines, efficacy, safety, and patient-centric factors such as quality of life and ease of administration.

The use of targeted therapies is supported by the efficacy demonstrated in randomized or single-arm trials in patients with corresponding alterations, real-world studies, and, for some agents, indirect treatment comparisons versus chemotherapy and/or immunotherapy.

Furthermore, targeted therapies have demonstrated generally favorable safety profiles, and are mostly orally administered, which may be perceived as a preferred option by some patients.

Clinical practice guidelines typically recommend initial treatment with matched targeted therapies that are approved for first-line use.

DIGITAL FEATURES

This article is published with digital features, including podcast audio and an infographic, to facilitate understanding of the article. To view digital features for this article, go to <https://doi.org/10.6084/m9.figshare.28548869>.

PODCAST TRANSCRIPT

Christine Bestvina: Hello. My name is Christine Bestvina. I'm an Associate Professor at the University of Chicago, and welcome to our podcast entitled "Driving best practices throughout the treatment journey for patients with non-small cell lung cancer with actionable alterations: a podcast discussion." Today, along with my co-authors, we're going to discuss the recently expanded therapeutic landscape for patients with non-small cell lung cancer with tumors harboring actionable alterations. We're going to discuss biomarker-driven treatment options and in this podcast; our goal is to provide perspectives on how to sequence different therapeutic options. We will discuss different treatment algorithms, and how to specifically decide for each individual patient how to sequence these therapies. We'll try to touch on efficacy, safety, as well as patient-centric considerations. With that, I'd like to ask my co-authors to introduce themselves.

Chul Kim: Thank you for having me. My name is Chul Kim. I'm a thoracic medical oncologist and an Associate Professor of Medicine at Georgetown University.

Nathalie Daaboul: Thank you again also for having me. Nathalie Daaboul, I'm an Associate Professor at the Université de Sherbrooke in Canada, and I'm also a hematologist and medical oncologist specializing in thoracic oncology.

Christine Bestvina: It's a pleasure to be here with you both today for this discussion.

We're going to start by providing a bit of background on actionable alterations and advanced non-small cell lung cancer. There have been so many advances in molecular diagnostics over the past few decades that have described the genetic heterogeneity of non-small cell lung cancer with numerous genomic driver alterations being implicated in both development as well as progression of disease [1]. We now have a growing list of drivers that are clinically actionable, meaning that approved targeted therapies are available to inhibit these underlying mechanisms driving oncogenesis in multiple tumor types with corresponding alterations

[2]. Dr. Kim, would you please start by outlining some of the key actionable alterations that we observe in non-small cell lung cancer?

Chul Kim: Certainly, though there are variations based on region and patient characteristics, it is estimated that approximately 60% of adenocarcinomas and about 10% of squamous cell carcinomas harbor actionable genomic alterations [2, 3]. Currently, actionable alterations include *EGFR* mutations, *BRAF* V600E mutations, *ERBB2* mutations, *KRAS* G12C mutations, *MET* exon 14 skipping alterations, as well as fusions involving genes such as *ALK*, *RET*, and *NTRK* [4, 5]. Most recently, *NRG1* gene fusion joined the list [5]. For these alterations, we have approved therapies including tyrosine kinase inhibitors, small molecules, and antibody–drug conjugates [5–7]. So, it is essential to perform molecular profiling to determine suitable treatment options for patients.

Christine Bestvina: Thanks so much. And Dr. Daaboul, could you talk a little bit about the significance of broad biomarker testing in non-small cell lung cancer?

Nathalie Daaboul: Broad biomarker testing is an essential part of diagnostics in non-small cell lung cancer. It is very important to have it at workup irrespective of patient characteristics, such as age, gender, or smoking history [5]. It enables identification of candidates for targeted therapy [5], and has become an essential part of our work. However, it's also important to mention that there are some subgroups that can be identified just by patient characteristics [5, 8]. For example, smoking history is more prevalent in *KRAS* G12C mutations [8, 9].

Christine Bestvina: Now that we've discussed how we think about the molecular landscape in non-small cell lung cancer, I'd like to transition to talking about therapeutic options for patients and how we sequence some of these. Historically, for patients who have non-small cell lung cancer, we had platinum-based chemotherapy as the mainstay of frontline treatment in non-small cell lung cancer [5, 10, 11]. In the past decade, immunotherapy has become an essential treatment option for many of our patients in the frontline setting. Whether immunotherapy is given as a single agent or in combination with chemotherapy

often depends on PD-L1 expression. Certain physicians may choose to use combination chemo-immunotherapy, even for patients who are PD-L1 high, depending on tumor burden or other patient characteristics [5, 12, 13]. However, the treatment sequencing gets very complicated when you start to add to it some of these different actionable alterations, such as *ALK* rearrangements, *BRAF* V600E mutations, *EGFR* mutations, and the remainder described by my co-authors. And so now we have to make decisions about how to sequence these and what to put first. Dr. Daaboul, what guidelines do you think can help practitioners make decisions in deciding how to sequence therapies for patients with actionable alterations?

Nathalie Daaboul: Of course, there are a few guidelines that we follow depending on where we are in the world. I think for initial treatment, the NCCN [National Comprehensive Cancer Network] guidelines' preferred options are matched targeted therapies that are approved for first-line use, while chemotherapy and/or immunotherapy are listed as other recommended regimens or useful in certain circumstances for some drivers [5]. Of course, we have other guidelines. It could be the ASCO [American Society of Clinical Oncology] or the ESMO [European Society for Medical Oncology] guidelines, and both are pretty much comfortable saying that they recommend initiating treatment with targeted agents first in the first line, rather than chemotherapy and/or immunotherapy, again, for most of the actionable genomics in non-small cell lung cancer [4, 14]. If we have used, for example, targeted therapy first, we know that we can sequence with chemo and/or immunotherapy, depending on the situation [4, 5, 14]. That said, there are some alterations that only have targeted agents in the second line. For example, I can imagine that we're thinking here about *KRAS* G12C, with the inhibitors being approved in the second line or further [4, 5, 14]. And for example, *ERBB2* mutations as of now [5], we also have inhibitors in some parts of the world that are indicated as second-line therapy, and we know that these agents are moving forward in the first line and that's where clinical trials are at the moment [15]. Of course, I think there

are also sometimes patients that we discover [have] an actionable driver while they're already on chemotherapy or chemotherapy and/or immunotherapy, and NCCN guidelines have put it forward that they recommend that for many actionable alterations, we can even interrupt systemic therapy to discuss with the patient if an approved targeted therapy could be interesting for them [5].

Christine Bestvina: That was a wonderful summary of such a quickly evolving field. I think there's no question that among these multiple driver alterations, it can get confusing about what's approved frontline, what's approved second line, and if patients should receive either chemo-immunotherapy or immunotherapy upfront. But I think you did a wonderful job summarizing how we can incorporate guidelines into these decision-making processes, how guidelines can be helpful for all practitioners in referencing: "Is this a frontline treatment option?" "Is this a second-line treatment option?" But in general, for those drugs that have approval in the frontline, we usually want to use the targeted agent first for a number of reasons, including both efficacy and toxicity.

Nathalie Daaboul: If I may add, it's such a changing field as well. So of course, we need to go back to those guidelines very frequently as approvals do change, and reimbursement and access as well.

Christine Bestvina: Absolutely. As Dr. Kim, our colleague, had brought up, we've even gotten new approvals in the last 6 months for things like *NRG1* gene fusions. So even for those of us who only think about lung cancer all day, every day, it's still tough to keep up sometimes. And so, it's nice for all of us to be able to have those resources, as well as each other, to potentially phone a friend when we need to decide when making some of these treatment decisions.

So now I'd like to transition to talking about treatment sequencing for patients who have actionable alterations. Dr. Kim, can you talk a little bit more about treatment sequencing and how you make some of these decisions about how to sequence therapies for patients who have actionable alterations?

Chul Kim: Yes, this is a topic of extensive research and perhaps some controversy. The availability of various guideline-recommended treatment options really underscores the need to consider multiple factors to optimize the sequencing of treatment for patients with tumors harboring actionable alterations [10]. One factor that may impact treatment sequencing is any delay in biomarker testing, which may lead clinicians to initiate chemotherapy and/or immunotherapy while awaiting the results, especially in patients requiring urgent symptom control [4, 16]. However, some studies suggest that this practice may be associated with less favorable outcomes, emphasizing the importance of initiating first-line treatment after obtaining molecular profiling data whenever feasible [17, 18].

Other key considerations, I would say, include relative efficacy and safety in different treatment lines, as well as patient-centric factors such as quality of life and ease of administration. Sometimes in later treatment lines, the emergence of specific molecular resistance mechanisms may inform treatment choices, such as *EGFR* T790M after first-generation *EGFR* tyrosine kinase inhibitors therapy and *EGFR* C797S after osimertinib or third-generation *EGFR* tyrosine kinase inhibitor therapy [5, 19]. Lastly, treatment sequencing decisions should also take into consideration the decreasing proportion of patients receiving therapy in each successive treatment line [20], which means that patients may not ultimately receive therapies that are deferred for later use. To give you some examples, some studies estimated that approximately 30% of patients with *EGFR*-mutant non-small cell lung cancer receive only one line of therapy [21], which is a number that sort of gives a lot of thought, with even higher rates in certain oncogene-driven subtypes [22].

Christine Bestvina: You referenced some of the tension that we see in clinic when the molecular profiling isn't back yet and you have to make a decision about a cycle of chemotherapy versus waiting for the molecular profiling. This just happened to me in clinic this week, and it took me 45 min to convince the patient and the family that waiting was the right thing to do. Can you speak a little bit about your

experience in the clinic, and how do you talk to patients about that decision?

Chul Kim: Yeah, this is a tough decision for some patients with a high burden of disease, and there's not a lot of time to wait for molecular profiling. At our center, we typically perform both tissue-based and blood-based NGS [next-generation sequencing], and blood-based NGS, as we all know, comes back much faster than tissue-based NGS [23]. So a lot of times, I wait for at least the blood-based NGS to come back before I make a treatment decision. But there are rare situations where you have to start a treatment, and usually that's chemotherapy for me, because I do not want to start immunotherapy. One particular situation we sometimes encounter is that the molecular profiling later shows certain driver mutations, such as *EGFR*. Then you're in trouble, because sequencing immunotherapy followed by TKI [tyrosine kinase inhibitor] sometimes can lead to different side effects [24]. So as you mentioned, explaining these points to patients is crucial. But I think this really comes down to individual decision-making, considering all the factors that the patient has. But there's a great tension, and this is a certain struggle that we have.

Christine Bestvina: Thanks for sharing your experience.

So now let's transition to talking about something exciting, which is the efficacy of so many of these drugs. Certainly for me, this is part of why I decided to go into thoracic oncology, because the efficacy of these targeted therapies is so much better than what we have been able to historically offer our patients. Dr. Daaboul, can you talk a little bit about efficacy as it relates to different actionable alterations, and also how that plays into your treatment decisions about first-line therapy?

Nathalie Daaboul: So this is an area that I would say does not always create consensus, right? We know that efficacy for many targeted agents has been approved based on single-arm trials due to the difficulty, mostly in conducting randomized trials in patients with tumors with rare alterations and that clearly complicates treatment decisions, and patients do rely on physician guidance for this [10]. Single-arm trials, we know, lack control groups and

randomization. They can result in potential bias and limited comparisons versus current standard-of-care treatment options [25]. So real-world evidence datasets can help us, but of course, are not as strict as comparative data, but these trials are valuable in early-phase research and situations where randomized clinical trials would be either not feasible or even ethically challenging. We remember one trial that did cause a lot of discussion after being presented. That was the LIBRETTO-431 trial where they compared seliperatinib, an anti-RET medication, to a platinum-based chemotherapy with or without pembrolizumab that showed clear superiority of seliperatinib [26]. However, the comparison was clearly challenged because we had already known the clear efficacy of a targeted agent in that situation, and it is well known that many targeted therapies generally result in longer progression-free survival than standard chemotherapy alone [27, 28].

Christine Bestvina: Excellent. And let's dive into that a little bit more. Dr. Kim, can you talk about how the approvals that are based on single-arm trials? How do we use this information when deciding how to sequence treatment for our patients? We're all taught always not to do cross-trial comparisons as fellows. But how do you, in practice, make decisions based on this information?

Chul Kim: Right. Sometimes it's tough. In cases where these trials have included treatment-naïve and previously treated patients, the data have been used to support approvals for line-agnostic indications [10, 29]. These line-agnostic approvals have sparked discussions about the optimal place for these targeted therapies within the treatment landscape. While they allow patients earlier access to new therapies and enable clinicians to tailor decisions based on the patient's disease characteristics and medical history, approvals based on single-arm trials [10, 30, 31] may leave some clinicians uncertain about how early use of these treatments compares to other standard-of-care treatment options.

Christine Bestvina: Despite the limitations that we've just discussed, do we know how effective targeted therapies will compare with chemo-immunotherapy? This is for patients

who have actionable alterations. I think this is something that frequently comes up in some of the debates at different conferences. Dr. Daaboul, how do you think about this space?

Nathalie Daaboul: I think this also arises in our clinics every day. Of course, efficacy varies between different drivers and classes of targeted agents. So we always need to go back to the books and trials to give us a better sense of the efficacy. However, many trials of approved targeted therapies have generally shown better outcomes than are obtained with chemotherapy [32]. We have talked about some phase 3, and I think I'll delve into that a bit more. We do have randomized control trials for some, I would say, actionable genomic alterations. But that said, seliprecitinib has shown to significantly increase progression-free survival versus chemotherapy [26].

I think what we know most is probably in the *EGFR* space and the prospective evidence for immunotherapy in patients with tumors with actionable alterations is limited. I think it's even more the case for *EGFR*, because we did have a few studies, including one that shows that pembrolizumab plus chemotherapy did not improve progression-free survival or overall survival versus placebo plus chemotherapy in a phase 3 trial for patients with *EGFR*-mutated, metastatic, non-small cell lung cancer that had previously received an appropriate targeted agent [33]. And again, we know that immunotherapy alone in a phase 2 trial for that exact patient population, so *EGFR*-mutated, advanced non-small cell lung cancer, really gave little benefit [34]. These are, of course, *EGFR* and *RET*, but overall, we can say that retrospective datasets have shown modest or poor efficacy in tumors with various oncogenic drivers [35], and that it really encompasses *ALK*, *BRAF*, *EGFR*, *KRAS*, *MET*, *RET*, and many of them. It has really highlighted that immunotherapy for these patients is not very efficacious, and they report potentially a response rate of 19%, a median PFS [progression-free survival] of not even 3 months, and again, a median overall survival that is poor, about a year [35].

Christine Bestvina: Excellent. Thank you for summarizing some of the prospective data that we have in this field. I think this is also a

place where we can look to real-world evidence to supplement the clinical trial data that we have. For example, we have one real-world study that looked at 1135 patients with advanced non-small cell lung cancer that had a driver alteration in a large US clinical database. For patients who received the appropriate targeted therapies, they had a median overall survival of 18.6 months compared to patients who did not receive the targeted therapy for which they were eligible, who had an overall survival of 11.4 months [11]. Of course, real-world evidence isn't perfect. There are many criticisms that we all could come up with, but I think when all of this data is taken together from what we have in single-arm trials, prospective randomized trials, and real-world evidence, I think again, this is all very strongly encouraging, if not demanding, that many of these patients be treated with the appropriate targeted therapy upfront. Dr. Kim, what additional evidence can be considered when direct head-to-head comparisons are not feasible?

Chul Kim: Yeah, sure. Probably I can touch on two additional genomic alterations that we typically see in clinic. In the absence of head-to-head data for some targeted therapies, indirect treatment comparisons with chemo and/or immunotherapy can be insightful. For instance, matched indirect comparisons have suggested better outcomes with the *MET* inhibitor tepotinib compared with immunotherapy plus chemo or immunotherapy alone in the first-line treatment of *MET* exon 14 skipping non-small cell lung cancer, which was recently presented [36]. Another study in previously treated patients with *MET* exon 14 skipping-positive non-small cell lung cancer in the VISION trial showed outcomes with tepotinib were better than those achieved by the same patients with first-line chemotherapy and/or immunotherapy before enrollment, based on retrospective review [37]. Similarly, in *BRAF*-mutant advanced non-small cell lung cancer, dabrafenib plus trametinib showed a lower risk of death and longer median overall survival compared to platinum-based chemotherapies [38]. Acknowledging the caveats, of course, of these retrospective analyses, the data support earlier use of targeted therapy in general,

especially given the limited number of treatment lines received by many patients [20].

Christine Bestvina: Thank you.

Alright, so we've done a great job, I think, summarizing the efficacy data in this space. Let's now transition to safety. Dr. Kim, can you please provide an overview of the safety considerations for targeted therapies in patients who have actionable alterations in non-small cell lung cancer?

Chul Kim: Sure. This topic could be a separate discussion on its own. Probably we need another podcast. But to highlight the key points, targeted therapies typically have a favorable safety profile, with generally mild and manageable adverse events [32, 39]. The characteristic adverse event profiles differ between classes of targeted agents, which is not surprising. For instance, skin toxicities are associated with the EGFR inhibitors [40] because EGFR is well expressed in certain parts of the body, including the skin, neurologic symptoms with the NTRK inhibitors [41], and peripheral edema with MET inhibitors [42]. Because of these, when we use these agents, we need to monitor and manage specific toxicities that may arise with these agents.

Christine Bestvina: Perfect. And then Dr. Daaboul, can you talk a little bit about how safety profiles may be affected based on sequencing therapies in this space?

Nathalie Daaboul: So of course, safety is always very important, as we know well, and for some targeted therapies, toxicities can be worse when a patient has received prior immunotherapy. I think that the easiest example would be for patients, as Dr. Kim had mentioned, who are *EGFR*-mutant and have received, for example, immunotherapy first; then we realize they have *EGFR*, and we want to switch them to, of course, a targeted agent. Unfortunately, there are reports of quite severe toxicities, including enhancing immune-related toxicities such as pneumonitis, colitis, and hepatitis [24]. Similarly, in patients with *KRAS* G12C mutations, prior immunotherapy has been linked to an increased risk of hepatotoxicity with sotorasib treatment [43, 44]. Even to the point that we may need to do some washout between therapies to decrease the risk of this

toxicity. I think these examples may support also the earlier use of targeted therapies.

Christine Bestvina: Excellent. And, yeah, these provide some really challenging situations in clinic. I recently saw a patient who'd gotten osimertinib followed by chemo-immunotherapy from another provider and had CNS progression, and really wanted to re-challenge her with osimertinib because of the CNS penetration of osimertinib. However, I was really limited by potential for toxicity here, so it can really become a complicated situation to make the right decision for the right patient.

Next, let's try to specifically put the patient back at the center of the discussion. So, Dr. Kim, how do patient-centric considerations such as ease of administration and quality of life influence the choice of treatment for patients with non-small cell lung cancer with actionable alterations?

Chul Kim: Yeah, this is an important aspect to consider when we make a treatment choice with patients. To highlight some differences, chemotherapy and immunotherapy require regular clinic visits for intravenous administration. Most targeted therapy agents are orally administered, which can increase patient convenience and may be perceived as a preferred option by some patients and their families [32]. Clinicians and patients can consider various factors, including efficacy, dosing, convenience, cost, and adverse events when evaluating treatment options [45–47]. Consistent with the efficacy, ease of administration, there's evidence to suggest that targeted therapies for non-small cell lung cancer are associated with greater improvements in quality of life and symptom relief compared to chemotherapy [26, 48–50]. These are important factors beyond efficacy and safety to consider when we form treatment plans for patients.

Christine Bestvina: And so while there are so many advantages to using targeted therapies for patients with actionable alterations, we would be remiss not to discuss some of the limitations and challenges, just in setting up our listeners for some of the things they may encounter in the clinic with their patients. Dr. Daaboul, can you speak a bit about some of the limitations and challenges that you've encountered, and

how you may have been able to overcome some of these challenges?

Nathalie Daaboul: I think that's an important point. There are challenges and limitations in everything that we do in lung cancer. Of course, targeted therapies are not an exception. So yes, targeted therapies for patients with actionable alterations in non-small cell lung cancer are recommended. They also come with some limitations. I think the most understood limitation would be drug resistance, potential toxicity in certain patients, and tumor heterogeneity as well that can unfortunately reduce efficacy.

But that is also something that we see in other tumor types, in other subtypes of lung cancer, and I think that really highlights the importance of continuing research to better understand what's really going on, the efficacy, safety, and even accessibility of these targeted therapies [6, 51]. As we are saying, yes, we want to go first with targeted agents, and I think that's the way to go. The optimal treatment sequencing of targeted therapy with chemo and/or immunotherapy still remains an important practical consideration, and something that we need to carefully look at now and in the future. I think something that we should also point out is that NGS testing or a broad panel is not done everywhere. Turnover times can be quite long. Even the precision of the test is not the same everywhere, and still, some patients undergo single-gene testing only, and they may not receive results for all recommended biomarkers [52], which can result in a reduced chance of receiving even an appropriate targeted therapy, if any. So I think here, this discussion is also highlighting the importance of having broad biomarker testing for all patients with advanced non-small cell lung cancer [5].

Christine Bestvina: Absolutely, test, test, test. We can't treat any mutations that we don't know are there.

Nathalie Daaboul: Absolutely.

Christine Bestvina: Alright, and so we're going to finish now with a few different case studies. Each of us will present a patient case where we go through some of the efficacy data, and some of the toxicity data, that inclined us to choose a particular treatment option for a

patient. Dr. Daaboul, can you share your patient case with us first?

Nathalie Daaboul: So I'll be discussing a lady that I saw at the clinic a few months ago, a 75-year-old with little comorbidities that presented with, of course, non-small cell lung cancer and adenocarcinoma. She had a lung mass, some pulmonary nodules, and a small brain metastasis. She was doing well. She was ECOG PS [Eastern Cooperative Oncology Group Performance Status Scale] 1 and had little symptoms. We had done broad testing, it took a few weeks to come back, but because she did not have that many symptoms, we decided to wait for the NGS to come in. A few weeks after, we discovered that she had *MET* exon 14 skipping alteration, and that she was PD-L1 low, so 1 to 49%. We had a discussion then with the patient and her family to see what were the alternatives of treatment, of course. So access is always important, and *MET* exon 14 skipping target agents where I practice are available. There's only one available. It would be tepotinib. And so we discussed tepotinib first line versus a standard chemotherapy plus immunotherapy. She had no contraindications to any of those treatments and after careful review, she preferred, and it is what we had also stressed, she had an actionable alteration. She clearly had medication that was available that was reimbursed, as we said, access is key, and we had an oral therapy to offer. So she went with tepotinib, and the treatment is ongoing right now.

After 3 months of treatment, we have re-scanned. Actually, everything is shrinking. So even the brain metastases that were not radiated have become smaller on MRI imaging, and even the lung nodules, everything is better and she is, it's a happy story, in a way, she is doing well and continuing the medication as of today.

Christine Bestvina: Excellent. Thank you for sharing that.

In my own clinic, I recently had a 70-year-old female who had a moderate smoking history, 30 pack-years. She had metastatic disease to bone, as well as two large lung lesions, each over 3 cm, no brain metastasis, though, and biopsy confirmed non-small cell lung cancer, adenocarcinoma, with a PD-L1 of 40% as well as a *BRAF* V600E mutation. And so with this

particular patient, we discussed a few different options. With her PD-L1 of 40%, we discussed chemo-immunotherapy as being her systemic treatment option. But then we also discussed encorafenib and binimetinib. And so we had data from the PHAROS study that was presented at ESMO 2024 [53], which to me was incredibly impressive from both an efficacy and a safety perspective. With encorafenib and binimetinib and longer-term follow-up, we saw progression-free survival of 30 months and a duration of response of 40 months, which is really impressive for non-small cell lung cancer.

This combination also is generally quite tolerable, and so we don't see the same amount of pyrexia that we've seen with earlier generations of BRAF-MEK inhibitor combinations. The patients generally experience nausea, diarrhea, fatigue, vomiting, which are symptoms that, in general, we deal with quite frequently in our clinic, and that with some aggressive management, whether it be independently or with some assistance from our palliative care colleagues, I think we all feel pretty comfortable managing. And so we, together, decided to pursue encorafenib and binimetinib as her frontline therapy. And similar to your patient, she is doing quite well. She is 15 months into treatment now with quite a robust response on imaging and tolerating the combination very well with just some antiemetic medication.

And last but not least, Dr. Kim, please share with us your case.

Chul Kim: My patient was a 52-year-old female who has not smoked, and she presented with abdominal girth, abdominal enlargement because of the high burden of disease in the liver. So during workup, she was found to have brain metastasis, and biopsy showed a large cell neuroendocrine carcinoma. Because of her initial symptoms, she was initiated on platinum–etoposide chemotherapy while awaiting molecular profiling results at another cancer center. Upon evaluation at our cancer center, molecular profiling came back and revealed a *RET* fusion. The patient didn't tolerate the chemotherapy too well, with grade 3 neutropenia and other side effects. And so after mutual discussion, we decided to try

selpercatinib, a *RET* inhibitor, and she achieved a durable response, both intracranially and extracranially, that lasted for several years. So I think this case really highlights the importance of molecular profiling that Dr. Daaboul and you, Christine, mentioned—the importance of comprehensive molecular testing in even non-adenocarcinoma subtypes of non-small cell lung cancer such as large cell neuroendocrine carcinoma, to guide optimal treatment selection.

Christine Bestvina: Excellent. Thank you both for sharing your recent patient experiences.

And so with that, we will go ahead and close out this podcast. Thank you so much to all of the listeners as we explored the treatment journey for patients with non-small cell lung cancer with actionable alterations. My co-authors and co-presenters did a wonderful job discussing the treatment landscape for these patients, as well as how we make clinical decisions for sequencing based on available treatments, how we incorporate both efficacy and safety into these discussions, and how a patient-centric approach can drive decisions for how we use targeted therapies for patients with actionable genomic alterations. Thank you so much for listening.

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